



**NMAM INSTITUTE
OF TECHNOLOGY**

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DATA CENTER

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Definition:

- A **data center** is a building, dedicated space within a building, or a group of buildings used to house computer systems and associated components, such as telecommunications and storage systems.
- Data center is a physical facility that organizations use to house their critical applications and data.
- A data center's design is based on a network of computing and storage resources that enable the delivery of shared applications and data.
- The key components of a data center design include routers, switches, firewalls, storage systems, servers, and application-delivery controllers.

Components of Data Centre

- ▶ **Network infrastructure:** This connects servers (physical and virtualized), data center services, storage, and external connectivity to end-user locations.
- ▶ **Storage infrastructure:** Data is the fuel of the modern data center. Storage systems are used to hold this valuable commodity.
- ▶ **Computing resources:** Applications are the engines of a data center. These servers provide the processing, memory, local storage, and network connectivity that drive applications.

Data Centre Architecture

Multitier Model

- ▶ The multitier model is the most common design in the enterprise.
- ▶ It is based on the web, application, and database layered design supporting commerce and enterprise business
- ▶ This type of design supports many web service architectures, such as those based on Microsoft .NET or Java 2 Enterprise Edition.

Data Centre Architecture

Multitier Model

- ▶ The multitier model uses software that runs as separate processes on the same machine using interprocess communication (IPC), or on different machines with communications over the network.
- ▶ Typically, the following three tiers are used:
 1. Web server
 2. Application
 3. Database

Data Centre Architecture

The server cluster

- ▶ The server cluster model has grown out of the university and scientific community to emerge across enterprise business verticals including financial, manufacturing, and entertainment.
- ▶ The server cluster model is most commonly associated with high-performance computing (HPC), parallel computing, and high-throughput computing (HTC) environments but can also be associated with grid/utility computing.
- ▶ These designs are typically based on customized, and sometimes proprietary, application architectures that are built to serve particular business objectives.

Features of Data Centre

1. Size: This would require a minimum area of around 5,000 to 30,000 square meters.

Example of Data Center of Google



Features of Data Centre

2. Data Security :

Another issue critical for data centers are the need to ensure maximum data security and 100 per cent availability.



3. Continuous monitoring 24x7x365:

24x7x365 monitoring must be there in data centers for the security of the data.

4. Security :

Physical security and **systems security** are critical to operations. *The Physical security can be achieved through security guards, proximity card and PIN for door access, biometrics access & PIN for door access and 24 x 365 CCTV surveillance and recording.*

Features of Data Centre

5. Availability of Data :

The goal of a data center is to maximize the availability of data, and to minimize potential downtime.

6. Electrical and power systems :

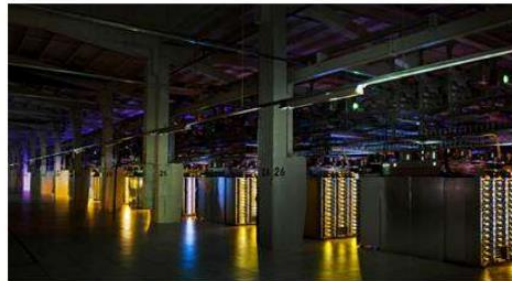
A data center should provide the highest power availability with uninterrupted power systems (UPS)

7. Backup Systems :

A data center should have provision of automatically providing backup data services to all shared hosting solutions.

8. Environment :

Cooling infrastructure is a significant part of a data center.



Benefits of Data Centre

- ▶ Following are the benefits or **advantages of Data Center**:
 - ➡ It offers services to the customers based on affordable rates in various plans as per lease period and requirements.
 - ➡ It offers robust software and hardware ecosystem.
 - ➡ It offers great system performance by distributing load across cluster nodes.
 - ➡ The users or customers need not have to worry about hiring people for maintenance in order to manage the data centers.
 - ➡ The data centers offer instant scalability based on changes in capacity requirements by customers.
 - ➡ The services are available always and without any failover due to backup of systems.

Challenges of Data Centre

1. Control skyrocketing data growth :

Data growth is the biggest data center hardware infrastructure challenge for large enterprises. However, there are several types of data deduplication technologies that helps in reducing data storage needs by eliminating redundant data.

2. System performance and scalability :

Many enterprises not only have had to add numerous disk-based backup target systems to stay ahead of growing data volumes but also to deal with an aging infrastructure or, in some cases, product obsolescence.

3. Network congestion and connectivity architecture :

The new generation of servers with multicore processors demands significantly high input/output (I/O), and if these servers are virtualized, this requirement further goes up, which is one of the biggest data center infrastructure challenge.

Challenges of Data Centre

4. IT administration and staff time at premium:

Data protection IT administrators have more data to protect and more complex data protection standards to meet while staying within a limited budget.

5. Inadequate Disaster Recovery plans:

Large enterprises that have been using physical tape backup systems & disparate disk-based solutions in branch offices are particularly vulnerable to downtime & data loss in the event of disaster.

5. Adopting new risk prone, cost-effective data protection technologies:

With limited budgets and resources, IT managers have worked to protect their investment in existing technologies.

7. Resource balancing:

The enterprise chief technical officer today needs to strike a working balance between reduced operational budgets, increased demands on existing infrastructure, maximizing availability, ensuring round-the-clock monitoring and management, and the periodic upgrades that today's technology demands.

Data center management Steps in transitioning to cloud- consolidation (1/3)

▶ 1. Consolidation

▶ **Objective:** Reduce physical infrastructure and improve efficiency.

▶ **Steps:**

▶ **Assessment:**

- ▶ Inventory all hardware, software, and services in the current data center.
- ▶ Identify redundant, outdated, or underutilized resources.
- ▶ Evaluate the performance, capacity, and utilization of existing assets.

Data center management Steps in transitioning to cloud- consolidation (2/3)

► **Planning:**

- Develop a consolidation strategy that aligns with business goals and compliance requirements.
- Create a detailed plan for migrating applications, data, and services to fewer, more efficient systems.
- Plan for physical consolidation if multiple data centers are being merged into a single location.

► **Virtualization:**

- Implement server virtualization to run multiple virtual machines (VMs) on a single physical server.
- Use storage virtualization to pool storage resources and improve utilization.

Data center management Steps in transitioning to cloud- consolidation (3/3)

▶ **Execution:**

- ▶ Migrate applications and data to the consolidated infrastructure.
- ▶ Decommission and dispose of obsolete hardware according to environmental and security guidelines.

▶ **Optimization:**

- ▶ Continuously monitor and optimize the consolidated environment to ensure it meets performance and capacity requirements.

Data center management Steps in transitioning to cloud- Automation (1/3)

- ▶ **Automation**

- ▶ **Objective:** Increase efficiency, reduce human error, and lower operational costs.

- ▶ **Steps:**

- ▶ **Identify Processes for Automation:**

- ▶ Catalog repetitive and time-consuming tasks suitable for automation.
- ▶ Prioritize tasks based on complexity, frequency, and impact on operations.

Data center management Steps in transitioning to cloud- Automation (2/3)

▶ **Choose Automation Tools:**

- ▶ Select appropriate automation tools and platforms (e.g., Ansible, Puppet, Chef) based on the organization's needs and existing infrastructure.

▶ **Develop Automation Scripts:**

- ▶ Create scripts to automate identified tasks, such as provisioning, configuration management, and software deployment.
- ▶ Ensure scripts are modular, reusable, and well-documented.

Data center management Steps in transitioning to cloud- Automation (3/3)

▶ **Test Automation Workflows:**

- ▶ Test automation scripts in a controlled environment to identify and resolve issues.
- ▶ Validate that automated processes meet security, compliance, and performance requirements.

▶ **Deploy and Monitor:**

- ▶ Implement automation in the production environment.
- ▶ Continuously monitor automated processes to ensure they function correctly and make adjustments as needed.

▶ **Continuous Improvement:**

- ▶ Regularly review and update automation workflows to address new requirements and improve efficiency.

IT as a Service (ITaaS)

- ▶ **Objective:** Transform IT from a cost center into a service-oriented organization that provides value to the business.
- ▶ **Steps:**
- ▶ **Define Service Catalog:**
 - ▶ Develop a catalog of IT services offered to the business, including infrastructure, platform, and software services.
 - ▶ Clearly define service levels, pricing, and delivery expectations for each service.

IT as a Service (ITaaS)

▶ **Implement Service Management:**

- ▶ Adopt a service management framework, such as ITIL (Information Technology Infrastructure Library), to guide the delivery and management of IT services.
- ▶ Establish processes for service request, incident management, change management, and service-level management.

▶ **Self-Service Portal:**

- ▶ Develop a self-service portal that allows users to request and manage IT services independently.
- ▶ Ensure the portal provides real-time visibility into service availability, performance, and usage.

IT as a Service (ITaaS)

▶ **Automation and Orchestration:**

- ▶ Integrate automation and orchestration tools to streamline the delivery and management of IT services.
- ▶ Enable automated provisioning, scaling, and decommissioning of services based on user requests and predefined policies.

▶ **Financial Management:**

- ▶ Implement chargeback or showback mechanisms to provide transparency into the cost of IT services.
- ▶ Align IT spending with business priorities and drive cost optimization.

▶ **Continuous Improvement:**

- ▶ Regularly review and refine the service catalog, service management processes, and self-service portal based on user feedback and changing business needs.
- ▶ Foster a culture of continuous improvement and innovation within the IT organization.